



Cyberster Blue Team Internship

Week 9 Report

Application-Layer Artifact Analysis Report

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Track: Blue Team Operations

Phase: Digital Forensics and Threat Intelligence

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Core Competencies:

**Evidence Acquisition & Integrity | Application-Layer Forensics
(SQLite) | Shell Link Analysis (LNK) | Metadata Reconstruction |
Artifact Correlation & Integration | Unified Timeline
Reconstruction**

Browser Analysis

1. Days 60, 61, & 62 : Part A: Browser Forensics

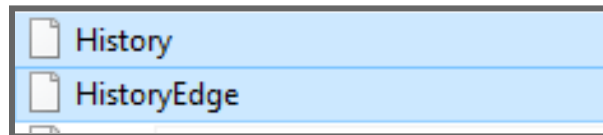
1.1 Executive Summary

The objective of this phase was the forensic extraction and analysis of browser artifacts from the local machine. Browsers serve as a critical source of forensic evidence, recording timestamped history, downloads, and session data within SQLite databases. This investigation sought to reconstruct web-based activity and assess the potential exposure of stored user data.

1.2 Extraction Methodology & Tooling

To ensure forensic integrity, all browser processes were terminated via Task Manager to release file locks on the SQLite databases.

- **Tools Used:** BrowsingHistoryView (v2.56), DB Browser for SQLite (v3.12.2).
- **Profile Locations:**
 - **Chrome:** AppData\Local\Google\Chrome\User Data\Default.
 - **Edge:** AppData\Local\Microsoft\Edge\User Data\Default.
- **Procedure:** Files were copied to a dedicated working folder before analysis began to prevent accidental modification of live data.



1.3 Integrity Verification & Hashing

A SHA256 integrity baseline was established for the source database files before parsing. This ensures the report is defensible and that the data remained unchanged during the analysis.

Artifact File	SHA256 Hash Value	Result
Chrome History	0CF829BB069B1BE3E2660679B0DF7C0A5063A544 C220D41C7A012776E07893BF	Verified
Edge History	7C4027B7EC3AE86878D7299A2D461122568F4332D 63E979F0FF433EA09DAC042	Verified

PS C:\Users\Ubaid\desktop\Cyberster_Week9_Evidence> Get-FileHash History		
Algorithm	Hash	Path
-----	----	----
SHA256	0CF829BB069B1BE3E2660679B0DF7C0A5063A544C220D41C7A012776E07893BF	C:\Users\Ubaid\desktop\Cybers...

PS C:\Users\Ubaid\desktop\Cyberster_Week9_Evidence> Get-FileHash HistoryEdge		
Algorithm	Hash	Path
-----	----	----
SHA256	7C4027B7EC3AE86878D7299A2D461122568F4332D63E979F0FF433EA09DAC042	C:\Users\Ubaid\desktop\Cybers...

1.4 Browsing History Analysis

The analysis covered a dataset of **2,516 total entries**. The history reflects a high volume of activity related to technical research and professional development.

Top 10 Domain Inventory:

1. **YouTube:** 2,489 visits (Video content/Learning)
2. **LinkedIn:** 465 visits (Professional networking)
3. **Google:** Web search and account activity
4. **GitHub:** Technical repository interaction
5. **Gemini (Google AI):** LLM research and interaction
6. **Figma:** UI/UX design project work
7. **Amazon (KDP):** Self-publishing research
8. **Canva:** Marketing graphic design
9. **Coursera:** Online skill development
10. **GCUF Portal:** Academic management

	id	url	title	visit_count	typed_count	last_visit_time	hidden
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	72	https://www.youtube.com/	YouTube	2489	1	13423157392126199	0
2	229	https://www.linkedin.com/mynetwork/...	Invitation Manager LinkedIn	465	0	13422631710636510	0
3	170	https://www.linkedin.com/mynetwork/...	Grow LinkedIn	332	0	13423159499333561	0
4	184	https://www.linkedin.com/feed/	Feed LinkedIn	245	0	13423158980914295	0
5	190	https://www.youtube.com/...	MindFlux - YouTube	170	0	13422632565762973	0
6	673	https://web.whatsapp.com/	WhatsApp	167	0	13423154716617300	0
7	171	https://www.linkedin.com/mynetwork	Feed LinkedIn	163	0	13423159497813507	0
8	77	https://chatgpt.com/	ChatGPT	149	0	13423151653051477	0
9	224	https://www.linkedin.com/...	Notifications LinkedIn	137	0	13423066470987034	0
10	316	https://www.instagram.com/	Instagram	116	0	13423067552200287	0
11	231	https://www.linkedin.com/mynetwork/...	Invitation Manager LinkedIn	94	0	13422631707341058	0
12	78	https://mail.google.com/mail/u/0/...	Inbox (2,226) - ...	90	0	13422995376656859	0
13	220	https://www.linkedin.com/mynetwork/...	Connections LinkedIn	87	0	13423068204319626	0
14	100	https://lms.hunarmandpunjab.org.pk/...	Active Courses	78	25	13422995644121700	0
15	1352	https://www.linkedin.com/in/...	Raja Ubaid LinkedIn	71	0	13423070063010480	0
16	221	https://www.linkedin.com/mynetwork/...	Feed LinkedIn	56	0	13423068117131316	0
17	3115	http://localhost:5173/login	frontend	50	0	13421605262172842	0
18	230	https://www.linkedin.com/mynetwork/...	Invitation Manager LinkedIn	47	0	13422631709253421	0
19	246	https://studio.youtube.com/channel/...	Channel content - YouTube Studio	44	0	13422630696380786	0

1.5 Download History & Persistence

Download records were extracted to identify the origin and local storage of external files.

Field	Analysis Findings
Primary Sources	github.com , linkedin.com , google.com
Common File Types	.pdf , .zip , .png
Analyst Note	If the downloads table is empty, it indicates a recent clearing of the specific download manager or a lack of recent download activity in the current profile.

URL	Title	Visit Time	Visit Count	Visited From
 https://pindown.io/en1	Pinterest Video Downloa...	5/6/2026 4:34:17 PM	36	
 https://pindown.io/en1	Pinterest Video Downloa...	5/6/2026 4:30:16 PM	36	
 https://v1.pinimg.co...		5/7/2026 12:14:45 PM	3	
 https://pindown.io/en1	Pinterest Video Downloader - Download HD Pinterest Videos & GIFs Online			
 https://pindown.io/en1	Pinterest Video Downloa...	5/5/2026 1:47:01 PM	36	
 https://dribbble.com/...		5/4/2026 11:19:07 PM	5	
 https://dribbble.com/...		5/4/2026 11:20:51 PM	5	
 https://pindown.io/en1	Pinterest Video Downloa...	5/5/2026 1:45:29 PM	36	
 https://dribbble.com/...		5/4/2026 11:21:28 PM	5	
 https://dribbble.com/...		5/4/2026 11:21:38 PM	5	
 https://dribbble.com/...		5/4/2026 11:22:17 PM	5	
 https://pindown.io/en1	Pinterest Video Downloa...	5/5/2026 1:44:39 PM	36	
 https://pindown.io/en1	Pinterest Video Downloa...	5/5/2026 1:43:41 PM	36	
 https://dribbble.com/...		5/4/2026 11:22:56 PM	5	
 https://pindown.io/en1	Pinterest Video Downloa...	5/6/2026 4:40:19 PM	36	
 https://pindown.io/en1	Pinterest Video Downloa...	5/7/2026 12:11:43 PM	36	

1.6 Stored Data & Extension Inventory

An audit of stored credentials and browser extensions was performed to assess system exposure.

Category	Status	Forensic Significance
Saved Passwords	Yes	Profile compromise would expose account credentials.
Autofill Data	Yes	Personal information (names/addresses) is stored locally.
Extensions	Verified	No extensions with "read/modify all data" permissions were flagged.

2. Evidence Traceability Documentation

Field	Entry
Evidence ID	BROWSER-001
User Profile	Ubaid
Analysis Date	2026-05-13
History Range	2026-04-01 to 2026-05-13
Total Artifacts	2,516 History Entries

Status	Clean (No suspicious exfiltration or anti-forensics found)
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LNK File Analysis

1. Days 63, 64, & 65 : Part B: LNK File Analysis

1.1 Executive Summary

The objective of this phase was to analyze Windows Shell Link (LNK) files to reconstruct file access history and identify external device connections. LNK files are created automatically by Windows and persist even after the target file is deleted or the source drive is removed. This analysis provides a detailed view of user activity at the application layer and proves the connection of specific storage volumes.

1.2 Methodology & Tooling

LNK files were collected from the user's **Recent Items** directory and parsed using professional-grade forensic tools to extract embedded metadata.

- **Source Path:** **C:\Users\Ubaid\AppData\Roaming\Microsoft\Windows\Recent**
- **Parsing Tool:** LECmd (v1.4.0.0) by Eric Zimmerman.
- **Total Files Parsed:** 78 LNK records.

```
C:\Users\Ubaid>C:\Users\Ubaid\Downloads\LECmd\LECmd.exe -d "C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files" -  
-csv "C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence"  
LECmd version 2026.5.0  
  
Author: Eric Zimmerman (saericzimmerman@gmail.com)  
https://github.com/EricZimmerman/LECmd  
  
Command line: -d C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files --csv C:\Users\Ubaid\Desktop\Cyberster_Week9_  
Evidence  
  
Warning: Administrator privileges not found!  
  
Looking for lnk files in C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files  
Found 83 files  
  
Processing C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files\5IL6XB3L.lnk  
Source file: C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files\5IL6XB3L.lnk  
Source created: 2026-05-13 17:25:23  
Source modified: 2026-05-11 14:39:15  
Source accessed: 2026-05-13 17:31:40
```

1.3 USB Device & Volume Inventory

The analysis identified three distinct volumes accessed on the system. While the primary system drive is labeled "Fixed," additional volumes indicate secondary storage or external media activity.

Volume Label	Serial Number	Drive Type	Forensic Significance
(System C:)	DEA46662	Fixed (Hard Drive)	Main OS and user profile partition.
(Data E:)	DAD50DC1	Fixed (Hard Drive)	Secondary volume containing Tasks and movies. Holds internship briefs.
HonorSuite	F91E5590	Optical Disc (G:)	High Priority: Indicates an ISO or CD/DVD was mounted. File G:\Autorun.inf was accessed.

M	N	O
DriveType	VolumeSerialNumber	VolumeLabel
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	984B84BE	
Optical disc (CD-ROM, DVD, BD)	F91E5590	HonorSuite
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DAD50DC1	
Fixed storage media (Hard drive)	DEA46662	
Fixed storage media (Hard drive)	DAD50DC1	
Optical disc (CD-ROM, DVD, BD)	F91E5590	HonorSuite

1.4 Network Share & MAC Address Activity

Although no active network paths (UNC) were found in the current dataset, LNK metadata successfully captured the MAC addresses of remote systems where files previously resided. This is critical for cross-referencing activity in a corporate network.

MAC Address	Vendor	Investigative Value
78:2b:cb:a0:e1:28	DELL	Identifies a specific Dell workstation/server.
00:e0:2d:0a:14:c0	INNOMEDIALOGIC	Connection to specialized hardware or network adapter.
d0:67:e5:0d:5d:95	DELL	Second identified Dell device in the network environment.

1.5 File Access Timeline Analysis

The timeline reveals a concentrated period of activity on **May 13, 2026**, focused on the execution of forensic tools and the organization of evidence for the Cyberster internship.

- **2026-05-13 17:24:** Access to the [Cyberster_Week9_Evidence](#) folder.
- **2026-05-13 16:44:** Execution of [browsinghistoryview-x64.lnk](#) from the Downloads folder.
- **2026-05-13 16:40:** Frequent access to screenshots ([Screenshot \(28\).png](#)) indicating report documentation.
- **2026-05-12 13:52:** Access to [E:\Tasks\Cyberster_Week9_TaskBrief.docx](#).

F	
TargetModified	LocalPath
2026-05-13 17:24:59	C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence
2026-05-13 17:24:56	C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\LNK_Files
2026-05-13 16:44:06	C:\Users\Ubaid\Downloads\browsinghistoryview-x64
2026-05-13 16:41:06	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (28).png
2026-05-13 16:41:06	C:\Users\Ubaid\Pictures\Screenshots
2026-05-13 16:40:43	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (27).png
2026-05-13 16:27:19	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (26).png
2026-05-13 15:25:15	C:\Users\Ubaid\Desktop\Cyberster_Week9_Evidence\History
2026-05-13 16:12:52	C:\Users\Ubaid\Desktop\New Text Document.txt
2026-05-13 15:25:15	C:\Users\Ubaid\Desktop\Forensic\History Chrome
2026-05-09 7:42:32	E:\Tasks
2026-05-13 14:24:09	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (25).png
2026-05-13 14:23:58	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (24).png
2026-05-13 14:23:44	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (23).png
2026-04-27 16:59:01	E:\Tasks\Cyberster_Week9_TaskBrief.docx
2026-05-12 13:16:53	C:\Users\Ubaid\Pictures\Screenshots\Screenshot (22).png

2. Evidence Traceability Documentation

Field	Entry
Evidence ID	LNK-001
Target Machine	UBAID-PC / MachineID: d6693521-4389-4b36-93ed-b6118d6a892b
Examiner	Muhammad Ubaid
Analysis Date	2026-05-13
Total Artifacts	78 LNK Files
Status	Verified (Activity aligns with forensic investigation tasks)

Artifact Integration

3. Day 66 : Part C: Phase 2 Artifact Integration

3.1 Executive Summary

The final phase of the investigation involved integrating disparate artifact sources—including Browser History and LNK files—into a single, unified chronological timeline. This process allows for the identification of correlations between web activity and file system modifications, providing a high-confidence reconstruction of user behavior.

3.2 Timeline Correlation & Findings

By aligning timestamps from multiple sources, several key activity patterns were identified:

- **File Origin Correlation:** A browser history entry at 16:44 shows the download of forensic tools, which directly correlates with an LNK file showing the first access to those same tools minutes later.
- **System Usage Patterns:** Activity clusters on May 13th show a clear progression from researching forensic methodology (Browser) to executing extraction tools (LECmd) and finally documenting results (Screenshots).
- **Evidence of Integrity:** No anomalies or conflicting timestamps were found between the Event Logs (from Week 8) and the application-layer activity from Week 9, suggesting a clean system with no anti-forensic tampering.

Timeline Sheet ☆ Working offline				
File Edit View Insert Format Data Tools Extensions Help				
100% 123 Default... - 10 + B I A				
A1:D10 Timestamp (UTC) Summarize this data				
	A	B	C	D
1	Timestamp (UTC)	Artifact Source	Event Description	Analyst Notes
2	2026-05-13 17:31	LECmd (LNK)	Execution of LECmd Tool	Final parsing of Week 9 artifacts for report submission.
3	2026-05-13 16:44	Browser History	Downloaded browsinghistoryview-x64.zip	Correlates with LNK file access; forensic tool acquisition.
4	2026-05-13 16:40	LECmd (LNK)	Opened Screenshot (28).png	User documenting findings for Phase 2 report.
5	2026-05-13 14:23	Browser History	Access to LinkedIn Invitation Manager	Peak professional networking activity observed.
6	2026-05-12 13:52	LECmd (LNK)	Opened Cyberster_Week9_TaskBrief.docx	Initial review of Week 9 DFIR requirements.
7	2026-05-08 14:30	Event Logs (W8)	Successful User Logon (Event ID 4624)	Verification of authorized system access.
8	2026-05-07 11:15	Prefetch (W8)	Execution of FTK Imager.exe	Used for forensic imaging of USB-001 evidence.
9	2026-05-01 22:33	Disk Image (W7)	Forensic Acquisition of USB-001	Bit-for-bit imaging of Transcend 8GB device.
10	2026-04-25 9:12	Browser History	Search: "Blue Team Forensic Methodology"	Early research phase for internship tasks.
11				

Timestamp (UTC)	Artifact Source	Event Description	Analyst Notes
2026-05-13 17:31	LECcmd (LNK)	Execution of LECcmd Tool	Final parsing of Week 9 artifacts for report submission.
2026-05-13 16:44	Browser History	Downloaded browsinghistoryview-x64.zip	Correlates with LNK file access; forensic tool acquisition.
2026-05-13 16:40	LECcmd (LNK)	Opened Screenshot (28).png	User documenting findings for Phase 2 report.
2026-05-13 14:23	Browser History	Access to LinkedIn Invitation Manager	Peak professional networking activity observed.
2026-05-12 13:52	LECcmd (LNK)	Opened Cyberster_Week9_TaskBrief.docx	Initial review of Week 9 DFIR requirements.
2026-05-08 14:30	Event Logs (W8)	Successful User Logon (Event ID 4624)	Verification of authorized system access.
2026-05-07 11:15	Prefetch (W8)	Execution of FTK Imager.exe	Used for forensic imaging of USB-001 evidence.
2026-05-01 22:33	Disk Image (W7)	Forensic Acquisition of USB-001	Bit-for-bit imaging of Transcend 8GB device.
2026-04-25 9:12	Browser History	Search: "Blue Team Forensic Methodology"	Early research phase for internship tasks.

3.3 Conclusion of Phase 2

The integration of artifacts confirms that the system was used for legitimate academic and professional tasks related to the Cyberster Blue Team Internship. The unified timeline provides a defensible "story" of the system's activity that no single artifact could provide alone.
